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Finite hypergeometric functions

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1 Introduction

Finite hypergeometric functions were introduced independently, and with different motivations, by John Greene [6] and Nick Katz [7, p.258] by the end of the 1980's. They are complex-valued functions on finite fields and an analogue of the classical analytic hypergeometric functions in one variable, also known as Thomae's functions. To display the analogy, we first recall the definition of the analytic hypergeometric functions.

Consider two multisets (sets with possibly repeating elements) $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_d)$ and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_d)$, where $\alpha_i, \beta_j \in \mathbb{Q}$ for all i, j. We assume for the moment that none of the α_i, β_j is in $\mathbb{Z}_{\leq 0}$ and $\beta_d = 1$. The hypergeometric function with parameter sets $\boldsymbol{\alpha}, \boldsymbol{\beta}$ is the analytic function defined by the power series in z

$$ {}_dF_{d-1}(\boldsymbol{\alpha}, \boldsymbol{\beta}|z) = \sum_{n=0}^{\infty} \frac{(\alpha_1)_n \cdots (\alpha_d)_n}{(\beta_1)_n \cdots (\beta_d)_n} \; z^n \; . $$

Here $(x)_n = x(x+1)\cdots(x+n-1)$ denotes the so-called Pochhammer symbol or rising factorial. Using the Γ-function we can rewrite the series as

$$ \frac{\Gamma(\beta_1)\cdots\Gamma(\beta_d)}{\Gamma(\alpha_1)\cdots\Gamma(\alpha_d)} \sum_{n=0}^{\infty} \frac{\Gamma(\alpha_1+n)\cdots\Gamma(\alpha_d+n)}{\Gamma(\beta_1+n)\cdots\Gamma(\beta_d+n)} \; z^n \; . $$

In order to display the similarity with the upcoming definition of finite hypergeometric functions, using the identity $\Gamma(x)\Gamma(1-x) = \pi/\sin(\pi x)$ we rewrite this as

$$ \sum_{n=0}^{\infty} \frac{\Gamma(\alpha_1+n)\cdots\Gamma(\alpha_d+n)\Gamma(1-\beta_1-n)\cdots\Gamma(1-\beta_d-n)}{\Gamma(\alpha_1)\cdots\Gamma(\alpha_d)\Gamma(1-\beta_1)\cdots\Gamma(1-\beta_d)} \; (-1)^{dn} z^n \; . $$

If β is a positive integer we interpret $\Gamma(1-\beta-n)/\Gamma(1-\beta)$ as $(-1)^n/(n+\beta-1)!$, the limit of $\Gamma(1-x-n)/\Gamma(1-x)$ as $x \to \beta$.

We now define finite hypergeometric functions. Again take two multisets $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, each consisting of d elements in $\mathbb{Q}$. Assume from now on that $\alpha_i \not\equiv \beta_j \pmod{\mathbb{Z}}$ for

all i, j. In the analytic case this condition is equivalent to the irreducibility of the hypergeometric differential equation. In the finite case we avoid certain degeneracies with this assumption. We need not assume $\beta_d = 1$ any more.
Let $\mathbb{F}_q$ be the finite field with q elements. Let ψ_q be a non-trivial additive character on $\mathbb{F}_q$ which we fix once and for all throughout this paper. For any multiplicative character $\chi : \mathbb{F}_q^\times \to \mathbb{C}^\times$ we define the Gauss sum

$$g(\chi) = \sum_{x \in \mathbb{F}_q^\times} \chi(x)\psi_q(x) \ .$$

Let ω be a generator of the character group on $\mathbb{F}_q^\times$ which we also fix throughout the paper. We use the notation $g(m) = g(\omega^m)$ for any $m \in \mathbb{Z}$. Note that $g(m)$ is periodic in m with period $q-1$. Very often we shall need characters on $\mathbb{F}_q^\times$ of a given order. For that we use the notation $\mathbb{1} = q-1$ so that a character of order d can be given by $\omega^{\mathbb{1}/d}$ for example, provided that d divides $\mathbb{1}$ of course.
Now we define finite hypergeometric sums. Let again $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ be multisets of d rational numbers each, and disjoint modulo $\mathbb{Z}$. Suppose in addition that q is such that

$$(q-1)\alpha_i, (q-1)\beta_j \in \mathbb{Z}$$

for all i and j.

Definition 1.1 (Finite hypergeometric sum) *Keep the above notation. We define for any $t \in \mathbb{F}_q$,*

$$H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t) = \frac{1}{1-q} \sum_{m=0}^{q-2} \frac{g(m+\alpha_1\mathbb{1}) \cdots g(m+\alpha_d\mathbb{1}) g(-m-\beta_1\mathbb{1}) \cdots g(-m-\beta_d\mathbb{1})}{g(\alpha_1\mathbb{1}) \cdots g(\alpha_d\mathbb{1}) g(-\beta_1\mathbb{1}) \cdots g(-\beta_d\mathbb{1})} \, \omega((-1)^d t)^m \ .$$

Note the analogy with the analytic hypergeometric function. These sums were considered without the normalizing factor $(\prod_{i=1}^d g(\alpha_i\mathbb{1})g(-\beta_i\mathbb{1}))^{-1}$ by Katz in [7, p258]. Greene, in [6], has a definition involving Jacobi sums which, after some elaboration, amounts to

$$\omega(-1)^{|\boldsymbol{\beta}|\mathbb{1}} q^{-d} \prod_{i=1}^d \frac{g(\alpha_i\mathbb{1})g(-\beta_i\mathbb{1})}{g(\alpha_i\mathbb{1} - \beta_i\mathbb{1})} \ H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t) \ ,$$

where $|\boldsymbol{\beta}| = \beta_1 + \cdots + \beta_d$. The normalization we adopt in this paper coincides with that of McCarthy, [9, Def 1.4].
It took some time before people realized that Greene's finite hypergeometric functions were closely related to point counting results on algebraic varieties over finite fields. As an example we mention the following result, adapted to our notation.

Theorem 1.2 (K. Ono, 1998) *Let q be an odd prime power and $\lambda \in \mathbb{F}_q$ and $\lambda \neq 0, 1$. Let E_λ be the projective elliptic curve given by the affine equation $y^2 = x(x-1)(x-\lambda)$ and $E_\lambda(\mathbb{F}_q)$ the set of $\mathbb{F}_q$-rational points (including infinity). Then*

$$|E_\lambda(\mathbb{F}_q)| = q + 1 - (-1)^{(q-1)/2} H_q(1/2, 1/2; 1, 1|\lambda) \ .$$

In this paper we propose a generalization of this theorem which applies to all hypergeometric sums which are *defined over* $\mathbb{Q}$. By that we mean that both $\prod_{j=1}^{d}(x-e^{2\pi i\alpha_j})$ and $\prod_{j=1}^{d}(x-e^{2\pi i\beta_j})$ are polynomials with coefficients in $\mathbb{Z}$. In other words, they are products of cyclotomic polynomials. Let us assume we are in this case. Then we can find natural numbers $p_1,\dots,p_r$ and $q_1,\dots,q_s$ such that

$$\prod_{j=1}^{d}\frac{x-e^{2\pi i\alpha_j}}{x-e^{2\pi i\beta_j}}=\frac{\prod_{j=1}^{r}x^{p_j}-1}{\prod_{j=1}^{s}x^{q_j}-1}\ .$$

Note that $p_1+\cdots+p_r=q_1+\cdots+q_s$. It is a small exercise to show that

$$\frac{(\alpha_1)_n\cdots(\alpha_d)_n}{(\beta_1)_n\cdots(\beta_d)_n}=M^{-n}\frac{(p_1n)!\cdots(p_rn)!}{(q_1n)!\cdots(q_sn)!},\quad M=\frac{p_1^{p_1}\cdots p_r^{p_r}}{q_1^{q_1}\cdots q_s^{q_s}}\ .$$

It turns out that when the hypergeometric parameters are defined over $\mathbb{Q}$, it is possible to extend the definition of H_q to all prime powers q which are relatively prime to the common denominator of the α_i,β_j. Let $D(X)$ be the greatest common divisor of the polynomials $\prod_{i=1}^{r}(X^{p_i}-1)$ and $\prod_{j=1}^{s}(X^{q_j}-1)$. We rewrite our Katz sum in a different shape.

Theorem 1.3 *With the above notation we have*

$$H_q(\boldsymbol{\alpha},\boldsymbol{\beta}|t)=\frac{(-1)^{r+s}}{1-q}\sum_{m=0}^{q-2}q^{-s(0)+s(m)}g(p_1m)\cdots g(p_rm)g(-q_1m)\cdots g(-q_sm)\,\omega(\epsilon M^{-1}t)^m\ ,$$

where $\epsilon=(-1)^{\sum_i q_i}$ *and* $M=\prod_{j=1}^{r}p_j^{p_j}\prod_{j=1}^{s}q_j^{-q_j}$, *and* $s(m)$ *is the multiplicity of the zero* $e^{2\pi im/\mathbb{1}}$ *in* $D(X)$.

This theorem is proven in Section 4.

Assumption 1.4 *From now on, when we work over* $\mathbb{Q}$, *we adopt the right-hand side of Theorem 1.3 as definition of* H_q.

Let $\lambda\in\mathbb{F}_q^\times$ and let V_λ be the affine variety defined by the projective equations

$$x_1+x_2+\cdots+x_r-y_1-\cdots-y_s=0,\qquad \lambda x_1^{p_1}\cdots x_r^{p_r}=y_1^{q_1}\cdots y_s^{q_s}$$

and $x_j,y_j\neq 0$. The main theorem of this paper reads as follows.

Theorem 1.5 *Let the notation for* p_i,q_j,M *be as above. Suppose that the greatest common divisor of* $p_1,\dots,p_r,q_1,\dots,q_s$ *is one and suppose that* $M\lambda\neq 1$. *Then there exists a suitable non-singular completion of* V_λ, *denoted by* $\overline{V_\lambda}$, *such that*

$$|\overline{V}_\lambda(\mathbb{F}_q)|=P_{rs}(q)+(-1)^{r+s-1}q^{\min(r-1,s-1)}H_q(\boldsymbol{\alpha},\boldsymbol{\beta}|M\lambda)\ ,$$

where

$$P_{rs}(q)=\sum_{m=0}^{\min(r-1,s-1)}\binom{r-1}{m}\binom{s-1}{m}\frac{q^{r+s-m-2}-q^m}{q-1}$$

and $\overline{V_\lambda}(\mathbb{F}_q)$ *is the set of* $\mathbb{F}_q$*-rational points on* $\overline{V_\lambda}$.

What we mean by a suitable non-singular completion of $\overline{V_\lambda}$ is elaborated in Section 5.
Here we give a few examples to illustrate the theorem.

Corollary 1.6 *Let $f(x) = x^3 + 3x^2 - 4t$ with $t \in \mathbb{F}_q$ and $t \neq 0, 1$. Let $N_f(t)$ be the number of zeros of $f(x)$ in $\mathbb{F}_q$. Suppose that q is not divisible by 2 or 3. Then*

$$N_f(t) = 1 + H_q(1/3, 2/3; 1, 1/2|t) \; .$$

Proof : We take $\alpha_1 = 1/3, \alpha_2 = 2/3, \beta_1 = 1, \beta_2 = 1/2$ and note that

$$\frac{(x - e^{2\pi i/3})(x - e^{4\pi i/3})}{(x-1)(x+1)} = \frac{x^3 - 1}{(x-1)(x^2-1)} \; .$$

So $p_1 = 3,\ q_1 = 1,\ q_2 = 2$, and $M = 27/4$. The variety V_λ is given by the equations $x - y_1 - y_2 = 0, \lambda x^3 = y_1 y_2^2$. Eliminate y_1 and set $y_2 = 1$ (dehomogenization) to get $x - \lambda x^3 - 1 = 0$. Replace x by $-3/x$ to get $x^3 + 3x^2 - 27\lambda = 0$. Application of Theorem 1.5 and the relation $t = 27\lambda/4$ gives the desired result.

□

Corollary 1.7 *Consider the elliptic curve $y^2 = x^3 - ax + b$ with $a, b \in \mathbb{F}_q$ and $a, b, -4a^3 + 27b^2 \neq 0$. Denote its completion with the point at infinity by $E_{a,b}$. Suppose that q is not divisible by 2 or 3. Then*

$$|E_{a,b}(\mathbb{F}_q)| = q + 1 - \omega(a/b)^{\mathbb{1}/2} q H_q(1/6, 5/6; 1/4, 3/4|27b^2/4a^3) \; .$$

Note that $27b^2/4a^3 = 1 - 1/J$ in terms of the J-invariant of the elliptic curve.

This is a reformulation of Theorem 2.1 in C. Lennon's paper [8]. To find it we need to replace Greene's hypergeometric sums with our H_q and apply the functional equation in Theorem 3.4, which is proved later in this paper. Note that this formula expresses counting results for any elliptic curve in terms of finite hypergeometric functions. Furthermore, $|qH_q(1/6, 5/6; 1/4, 3/4|t)| \leq 2\sqrt{q}$ so the values of H_q are not integers in this case.

Proof : We take $\alpha_1 = 1/6, \alpha_2 = 5/6, \beta_1 = 1/4, \beta_2 = 3/4$ and note that

$$\frac{(x - e^{2\pi i/6})(x - e^{10\pi i/6})}{(x - e^{2\pi i/4})(x + e^{6\pi i/4})} = \frac{(x^6 - 1)(x - 1)}{(x^3 - 1)(x^4 - 1)} \; .$$

So $p_1 = 6,\ p_2 = 1,\ q_1 = 3,\ q_2 = 4$, and $M = 27/4$. The variety V_λ is given by the equations

$$x_1 + x_2 - y_1 - y_2 = 0, \lambda x_1^6 x_2 = y_1^3 y_2^4 \; .$$

Eliminate x_2 and set $x_1 = 1$ to dehomogenize. We get

$$1 + \lambda^{-1} y_1^3 y_2^4 - y_1 - y_2 = 0 \; .$$

Introduce new coordinates x, y via

$$x = \frac{1 - y_2}{y_1 y_2^2}, \qquad y = \frac{1 - \lambda^{-1} y_1^2 y_2^4 - y_2}{y_1 y_2^3} \ .$$

Then we get the curve

$$y^2 = x^3 - \lambda^{-1} x + \lambda^{-1} \ .$$

Theorem 1.5 tells us that the number of $\mathbb{F}_q$-rational points (including ∞) equals

$$q + 1 + q H_q(1/6, 5/6; 1/4, 3/4 | 27\lambda/4) \ .$$

Now we start with the curve $y^2 = x^3 + ax + b$ and transform it to the above curve with parameter λ. Replace x by bx/a to get

$$\frac{a^3}{b^3} y^2 = x^3 - \frac{a^3}{b^2} x + \frac{a^3}{b^2}$$

which is our curve with $\lambda = b^2/a^3$ and a quadratic twist by a/b. So we find that the number of $\mathbb{F}_q$-rational points on $y^2 = x^3 - ax + b$ is given by

$$q + 1 + \omega (a/b)^{\mathbb{1}/2} q H_q(1/6, 5/6; 1/4, 3/4 | 27 b^2/4a^3) \ .$$

□

Corollary 1.8 *Consider the rational elliptic surface S_λ given by the affine equation*

$$y^2 - xyz + x^3 + z^5 - \lambda^{-1} = 0$$

with $\lambda \in \mathbb{F}_q^\times$. Suppose that q is not divisible by $2, 3$ or 5 and $2^{14} 3^9 5^5 \lambda \neq 1$. Let $\overline{S_\lambda}$ be a suitable completion of S_λ. Then

$$|\overline{S_\lambda}(\mathbb{F}_q)| = q^2 + 3q + 1 + q H_q(\boldsymbol{\alpha}, \boldsymbol{\beta} | 2^{14} 3^9 5^5 \lambda) \ ,$$

where

$$\boldsymbol{\alpha} = (1/30, 7/30, 11/30, 13/30, 19/30, 23/30, 29/30)$$

and

$$\boldsymbol{\beta} = (1/5, 1/3, 2/5, 1/2, 3/5, 2/3, 4/5, 1) \ .$$

Moreover, $H_q(\boldsymbol{\alpha}, \boldsymbol{\beta} | t)$ is integer valued.

From [1] it follows that the analytic function ${}_8F_7(\boldsymbol{\alpha}, \boldsymbol{\beta} | z)$ is an algebraic function. Fernando Rodriguez-Villegas computed its degree over $\mathbb{C}(z)$ [4], which is 483840. He also noted that

$${}_8F_7(\boldsymbol{\alpha}, \boldsymbol{\beta} | 2^{14} 3^9 5^5 t) = \sum_{n \geq 0} \frac{(30n)! n!}{(15n)! (10n)! (6n)!} \ t^n \ .$$

The coefficients turn out to be integers and they were essentially used by Chebyshev in his proof for his estimates of the prime counting function $\pi(x)$.
Another remark is that $y^2 - xyz + x^3 + z^5 - \lambda^{-1} = 0$ is a rational elliptic surface for any given λ. The ζ-function of such surfaces has been computed extensively by Shioda in[11]. There it turns out that the global ζ-function of such a surface has the form $\zeta(s)\zeta(s-1)^2\zeta(s-2)L(\rho, s-1)$, where $\zeta(s)$ is the Riemann ζ function and $L(\rho, s)$ is the Artin L-series corresponding to a finite representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ of dimension 8.

Proof : We take the elements of $\boldsymbol{\alpha}$ for the α_i and the elements of $\boldsymbol{\beta}$ for the β_j. We verify that

$$\prod_{j=1}^{8} \frac{x - e^{2\pi\alpha_j}}{x - e^{2\pi i\beta_j}} = \frac{(x^{30}-1)(x-1)}{(x^{15}-1)(x^{10}-1)(x^6-1)} \ .$$

Compare the exponents with the remarks above. We have $M = 2^{14}3^95^5$. Theorem 1.5 implies that

$$q^2 + 3q + 1 - qH_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|M\lambda)$$

equals the number of $\mathbb{F}_q$-rational points on a suitable completion of

$$x_1 + x_2 - y_1 - y_2 - y_3 = 0, \qquad \lambda x_1^{30}x_2 = y_1^{15}y_2^{10}y_3^6 \ .$$

Eliminate x_2 and set $x_1 = 1$. We obtain

$$1 + \lambda^{-1}y_1^{15}y_2^{10}y_3^6 - y_1 - y_2 - y_3 = 0 \ .$$

Substitute $y_1 \to y/(xz), y_2 \to x^2/(yz), y_3 \to z^4/(xy)$ to obtain

$$-xyz - \lambda^{-1} + y^2 + x^3 + z^5 = 0 \ .$$

The integrality of the values of H_q follows from Theorem 4.2.

□

Surprisingly enough Theorem 1.5 does not immediately imply Ono's Theorem 1.2. In this case we get the parameters $p_1 = p_2 = 2, q_1 = q_2 = q^3 = q^4 = 1$ and the threefold

$$x_1 + x_2 - y_1 - y_2 - y_3 - y_4 = 0, \qquad \lambda x_1^2x_2^2 = y_1y_2y_3y_4 \ ,$$

instead of the expected Legendre family of elliptic curves. However in this case we have the relations $p_1 = q_1 + q_2, p_2 = q_3 + q_4$ besides the overall relation $p_1 + p_2 = q_1 + \cdots + q_4$. In such a case we can construct another variety whose point count also yields the desired hypergeometric sum. We indicate how this works in Theorem 6.1 in the last section of this paper. This theorem does yield Legendre's family for our example.
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reverse engineering hypergeometric motives on an industrial scale arose. We thank both AIM and ICTP for their hospitality. In addition we would like to thank the Mathematisches Forschungsinstitut Oberwolfach for providing the wonderful atmosphere during the workshop 'Explicit methods in number theory' in July 2013. Here the final form of the present paper was conceived.

2 Gauss sums, basic properties

We use the notation given in the introduction and recall some basic theorems. The reference that we use is the second author's books on Number Theory, [2], [3] (where the notation $\tau(\chi, \psi)$ is used instead of $g(\chi)$ though).

Theorem 2.1 *We have*

1. $g(0) = -1$
2. $g(m)g(-m) = \omega(-1)^m q$ *if* $m \not\equiv 0 \pmod{\mathbb{1}}$.

This is Lemma 2.5.8 and Proposition 2.5.9 of [2].

Theorem 2.2 *Define for any two integers m, n the Jacobi sum*

$$J(m,n) = \frac{g(m)g(n)}{g(n+m)}\,.$$

Then,

$$J(m,n) = \begin{cases} \sum_{x \in \mathbb{F}_q \setminus \{0,1\}} \omega(1-x)^m \omega(x)^n & \text{if } m, n, m+n \not\equiv 0 \pmod{\mathbb{1}} \\ -1 & \text{if } m \equiv 0 \pmod{\mathbb{1}} \text{ or } n \equiv 0 \pmod{\mathbb{1}} \\ \omega(-1)^m q & \text{if } m \equiv -n \pmod{\mathbb{1}} \text{ and } m \not\equiv 0 \pmod{\mathbb{1}} \end{cases}$$

This follows from Corollary 2.5.17 of [2], although we slightly perturbed the definition of J. A simple consequence is the following.

Lemma 2.3 *For every pair of integers m, n the quotient $g(m)g(n)/g(m+n)$ is an algebraic integer in $\mathbb{Q}(\zeta_{\mathbb{1}})$, where $\zeta_{\mathbb{1}}$ is a primitive $q-1$-st root of unity.*

In Theorems 3.3 and 4.2 (in the case over $\mathbb{Q}$), we will compute a bound for the denominator of H_q. To do so we need a statement on the prime ideal factorization of Gauss sums. This is provided by Stickelberger's theorem. We use Theorem 3.6.6 of [2], in a weaker form.

Theorem 2.4 (Stickelberger) *Let $\mathfrak{p}$ be the ideal divisor of p in $\mathbb{Z}[\zeta_{\mathbb{1}}]$ such that $\omega^{-1}(x) \equiv x \pmod{\mathfrak{p}}$ for all $x \in \mathbb{F}_q$. Note that $\mathfrak{p}$ has degree f, where $p^f = q$. Let $\mathfrak{P}$ be the totally ramified prime ideal above in $\mathfrak{p}$ in $\mathbb{Z}[\zeta_{\mathbb{1}}, \zeta_p]$. Let $0 \leq r < q-1$. Then*

$g(r)$ is exactly divisible by $\mathfrak{P}^{\sigma(r)}$, where $\sigma(r)$ is the sum of the digits r_i in the base p decomposition $r = r_0 + r_1 p + \cdots + r_{f-1}p^{f-1}$.
An alternative description of $\sigma(r)$, without the restriction $0 \le r < q-1$, is given by

$$\sigma(r) = (p-1)\sum_{i=1}^{f} \left\{ \frac{p^i r}{q-1} \right\} .$$

In order to rewrite H_q when it is defined over $\mathbb{Q}$ we use the following result.

Theorem 2.5 (Hasse-Davenport) *For any $N \in \mathbb{N}$ dividing $\mathbb{1}$ we have*

$$g(Nm) = -\omega(N)^{Nm} \prod_{j=0}^{N-1} \frac{g(m + j\mathbb{1}/N)}{g(j\mathbb{1}/N)} .$$

This is Theorem 3.7.3 of [2]. Note the analogy with Euler's identity

$$\Gamma(Nz) = N^{Nz} \frac{\prod_{j=0}^{N-1} \Gamma(z + j/N)}{\prod_{j=1}^{N-1} \Gamma(j/N)}$$

for the Γ-function. The following proposition will be used repeatedly. It is Fourier inversion on the multiplicative characters.

Proposition 2.6 *Let $G : \mathbb{F}_q^\times \to \mathbb{C}$ be a function. Then, for any $\lambda \in \mathbb{F}_q^\times$ we have*

$$G(\lambda) = \frac{1}{q-1} \sum_{m=0}^{q-2} G_m \omega(\lambda)^m ,$$

where

$$G_m = \sum_{\lambda \in \mathbb{F}_q^\times} G(\lambda)\omega(\lambda)^{-m} .$$

Here is an application for later use.

Lemma 2.7 *Let $\mathbf{a} = (a_1, \ldots, a_n) \in \mathbb{Z}^n$. Define $a_{n+1} = -a_1 - \cdots - a_n$ and $a = \gcd(a_1, \ldots, a_n)$. Then, for any integer m,*

$$\sum_{v \in \mathbb{F}_q, \mathbf{x} \in (\mathbb{F}_q^\times)^n} \psi_q(v(1 + x_1 + \cdots + x_n))\omega(\mathbf{x}^{\mathbf{a}})^m = (q-1)^n \delta(am) + g(a_1 m) \cdots g(a_{n+1} m) ,$$

where $\mathbf{x} = (x_1, \ldots, x_n)$, $\mathbf{x}^{\mathbf{a}} = x_1^{a_1} \cdots x_n^{a_n}$, and $\delta(x) = 1$ if $x \equiv 0 \pmod{\mathbb{1}}$ and 0 otherwise.

Proof : We carry out the summation over $x_1, \ldots, x_n$ and $v = 0$ to get

$$\sum_{\mathbf{x} \in (\mathbb{F}_q^\times)^l} \omega(\mathbf{x}^{\mathbf{a}})^m = (q-1)^n \prod_{i=1}^{n} \delta(a_i m) = (q-1)^n \delta(am) \ .$$

The summation over $x_1, \ldots, x_n$ and $v \in \mathbb{F}_q^\times$ yields

$$\begin{aligned} \sum_{v, x_i \in \mathbb{F}_q^\times} \psi_q(v(1 + x_1 + \cdots + x_n)) \omega(\mathbf{x}^{\mathbf{b}})^m &= \sum_{v \in \mathbb{F}_q^\times} g(a_1 m) \cdots g(a_n m) \psi_q(v) \omega(v^{(-a_1 - \cdots - a_n)m}) \\ &= g(a_1 m) \cdots g(a_{n+1} m). \end{aligned}$$

From these two summations our result follows.

□

3 Katz hypergeometric functions

Let $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_d)$ and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_d)$ be disjoint multisets in $\mathbb{Q}/\mathbb{Z}$. We assume that both $(q-1)\boldsymbol{\alpha}$ and $(q-1)\boldsymbol{\beta}$ are subsets of $\mathbb{Z}$. In [7, p 258] Katz considered for any $t \in \mathbb{F}_q^\times$ the exponential sum

$$Hyp_q(\boldsymbol{\alpha}, \boldsymbol{\beta} | t) = \sum_{(\mathbf{x}, \mathbf{y}) \in T_t} \psi_q(x_1 + \cdots + x_d - y_1 - \cdots - y_d) \ \omega(\mathbf{x})^{\boldsymbol{\alpha}\mathbb{1}} \omega(\mathbf{y})^{-\boldsymbol{\beta}\mathbb{1}} \ ,$$

where T_t is the toric variety defined by $t x_1 \cdots x_d = y_1 \cdots y_d$ and $\omega(\mathbf{x})^{\boldsymbol{\alpha}\mathbb{1}}$ stands for $\omega(x_1)^{\alpha_1 \mathbb{1}} \cdots \omega(x_d)^{\alpha_d \mathbb{1}}$, etc. Note that we have taken Katz's formula for the case $n = m = d$. Define

$$S_q(\boldsymbol{\alpha}, \boldsymbol{\beta} | t) = \frac{1}{q-1} \sum_{m=0}^{q-2} \prod_{i=1}^{d} \ g(m + \alpha_i \mathbb{1}) g(-m - \beta_i \mathbb{1}) \ \omega((-1)^d t)^m \ .$$

Proposition 3.1 (Katz) *With the above notation we have*

$$Hyp_q(\boldsymbol{\alpha}, \boldsymbol{\beta} | t) = \omega(-1)^{|\boldsymbol{\beta}|\mathbb{1}} S_q(\boldsymbol{\alpha}, \boldsymbol{\beta} | t)$$

for all $t \in \mathbb{F}_q^\times$, where $|\boldsymbol{\beta}| = \sum_i \beta_i$.

Proof : We consider $Hyp_q(\boldsymbol{\alpha}, \boldsymbol{\beta} | t)$ as function of t and determine the m-th coefficient of its Fourier expansion. Using Proposition 2.6, this reads

$$\frac{1}{q-1} \sum_{t, x_i, y_j \in \mathbb{F}_q^\times, t x_1 \cdots x_d = y_1 \cdots y_d} \psi_q(x_1 + \cdots + x_d - y_1 - \cdots - y_d) \ \omega(\mathbf{x})^{\boldsymbol{\alpha}\mathbb{1}} \omega(\mathbf{y})^{-\boldsymbol{\beta}\mathbb{1}} \omega(t)^{-m} \ .$$

We substitute $t = (y_1 \cdots y_d)(x_1 \cdots x_d)^{-1}$ to get

$$\frac{1}{q-1} \sum_{x_i, y_j \in \mathbb{F}_q^\times} \psi_q(x_1 + \cdots + x_d - y_1 - \cdots - y_d)\ \omega(\mathbf{x})^{m+\boldsymbol{\alpha}\mathbb{1}} \omega(\mathbf{y})^{-m-\boldsymbol{\beta}\mathbb{1}} \ .$$

Summation over the $2d$ variables x_i, y_j then gives the desired result.

□

Notice that Hyp_q (and S_q) are algebraic integers in the field $\mathbb{Q}(\zeta_p, \zeta_{q-1})$, where ζ_n denotes a primitive n-th root of unity. A simple consideration shows that S_q is in $\mathbb{Q}(\zeta_{q-1})$ if $\sum_{i=1}^d (\alpha_i - \beta_i) \in \mathbb{Z}$. Moreover we have the following divisibility properties.

Proposition 3.2 *For any integer m the product $\prod_{i=1}^d g(m + \alpha_i \mathbb{1}) g(-m - \beta_i \mathbb{1})$ is divisible by $g(|\boldsymbol{\alpha} - \boldsymbol{\beta}|\mathbb{1})$, where $|\boldsymbol{\alpha} - \boldsymbol{\beta}| = \sum_{i=1}^d (\alpha_i - \beta_i)$, and by $\prod_{i=1}^d g((\alpha_i - \beta_i)\mathbb{1})$. Moreover, the quotients are in $\mathbb{Q}(\zeta_{q-1})$.*

This is a direct consequence of Lemma 2.3. Consequently $S_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t)$ is divisible by these numbers for any $t \in \mathbb{F}_q^\times$. This proposition suggests that we might normalize S_q by the factor $1/g(|\boldsymbol{\alpha} - \boldsymbol{\beta}|\mathbb{1})$ or by $1/\prod_i g((\alpha_i - \beta_i)\mathbb{1})$. The latter normalization is taken by John Greene in his definition of finite hypergeometric sums, see [6]. More precisely, Greene uses the function

$$\frac{\pm q^{1-d}}{\prod_i g((\alpha_i - \beta_i)\mathbb{1})} S_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t) \ .$$

We will adopt the normalization from the introduction. It keeps the symmetry in the α_i and β_j, but accepts the possibility that the function values may not be integers. Namely,

$$H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t) = -\prod_{i=1}^d \frac{1}{g(\alpha_i \mathbb{1}) g(-\beta_i \mathbb{1})} S_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t) \ .$$

To get a bound on the denominator of H_q we introduce the *Landau function*

$$\lambda(\boldsymbol{\alpha}, \boldsymbol{\beta}, x) = \sum_{i=1}^d \left(\{x + \alpha_i\} - \{\alpha_i\} + \{-x - \beta_i\} - \{-\beta_i\} \right) \ ,$$

where $\{x\}$ denotes the fractional part of the real number x.

Theorem 3.3 *Keep the above notation and let N be the common denominator of the α_i, β_j. Define*

$$\lambda = -\min_{x \in [0,1], \gcd(k,N)=1} \lambda(k\boldsymbol{\alpha}, k\boldsymbol{\beta}, x) \ .$$

Then $q^\lambda H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t)$ is an algebraic integer.

Proof : The definition of Hyp_q and Proposition 3.1 imply that $S_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t)$ is an algebraic integer. Since H_q is a normalization of S_q by a factor $\prod_{i=1}^{d} g(\alpha_i \mathbb{1}) g(\beta_i \mathbb{1})$ the denominator of H_q is a power of p. We determine the p-adic valuation of the coefficients of the expansion of H_q.
Let $\mathfrak{p}$ be defined as in Theorem 2.4. Then the $\mathfrak{p}$-adic valuation of the m-th coefficient of $H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t)$ equals

$$\sum_{j=1}^{d} \sum_{i=0}^{f-1} \left(\left\{ p^i \frac{m}{q-1} + \alpha_j \right\} + \left\{ p^i \frac{-m}{q-1} - \beta_j \right\} - \{\alpha_j\} - \{-\beta_j\} \right) .$$

Note that this is greater than or equal to $-f\lambda$. Let $\sigma \in \mathrm{Gal}(\zeta_{q-1})/\mathbb{Q})$ and k such that $\sigma(\zeta_{q-1}) = \zeta_{q-1}^k$. Then the $\sigma(\mathfrak{p})$-adic valuations of the coefficients of $H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t)$ are equal to the $\mathfrak{p}$-adic valuations of the coefficients of $H_q(k^*\boldsymbol{\alpha}, k^*\boldsymbol{\beta}|t)$, where $kk^* \equiv 1 \pmod{q-1}$. The latter valuations are also bounded below by $-f\lambda$, so our theorem follows.

□

We end by noting some obvious identities for S_q.

Theorem 3.4 *Let $\boldsymbol{\alpha}, \boldsymbol{\beta}$ be as before. Let $\mu \in \mathbb{Q}$ be such that $\mu \mathbb{1} \in \mathbb{Z}$ and denote $(\alpha_1 + \mu, \ldots, \alpha_d + \mu)$ by $\mu + (\alpha_1, \ldots, \alpha_d)$, etc. For any $\lambda \in \mathbb{F}_q^\times$ we have*

$$\omega(t)^{\mu \mathbb{1}} S_q(\mu + \boldsymbol{\alpha}, \mu + \boldsymbol{\beta}|t) = S_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t) .$$

This follows directly from the definition of S_q and the replacement of the summation variable m by $m - \mu \mathbb{1}$. As an example in the analytic setting we recall Gauss' analogues ${}_2F_1(\alpha, \beta, \gamma|z)$ and $z^{1-\gamma}\ {}_2F_1(\alpha + 1 - \gamma, \beta + 1 - \gamma, 2 - \gamma|z)$ (when γ is not an integer). In contrast with the finite case, the latter are linearly independent over the constant field. We also have

Theorem 3.5 *Let $\boldsymbol{\alpha}, \boldsymbol{\beta}$ be as before and suppose $t \in \mathbb{F}_q^\times$. Then*

$$S_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t) = S_q(-\boldsymbol{\beta}, -\boldsymbol{\alpha}|1/t) .$$

Again the proof follows directly from the definition of S_q and replacing the summation variable m by $-m$.

4 Hypergeometric motive over $\mathbb{Q}$

From now on we concentrate on the case when the Katz sums are defined over $\mathbb{Q}$. In other words, as in the introduction, we assume the following:

Assumption 4.1 *There exist natural numbers $p_1, \ldots, p_r, q_1, \ldots, q_s$ such that*

$$\prod_{j=1}^{d} \frac{X - e^{2\pi i \alpha_j}}{X - e^{2\pi i \beta_j}} = \frac{\prod_{i=1}^{r}(X^{p_i} - 1)}{\prod_{j=1}^{s}(X^{q_j} - 1)} \ .$$

We now prove Theorem 1.3, which tells us that we can rewrite our Katz sum in a different shape.

Proof of Theorem 1.3: We use the notation given in the introduction. Let δ be the degree of $D(X)$. Suppose that the zeros of $D(X)$ are given by $e^{2\pi i c_j/\mathbb{1}}$ with $j = 1, \ldots, \delta$, where possible repetitions of the roots are allowed. The coefficient of $\omega(t)^m$ (without the $1/(1-q)$) can be rewritten as

$$\omega(-1)^{dm} \left(\prod_{i=1}^{r} \prod_{j=0}^{p_i - 1} \frac{g(m + j\mathbb{1}/p_i)}{g(j\mathbb{1}/p_i)} \right) \left(\prod_{i=1}^{s} \prod_{j=0}^{q_i - 1} \frac{g(-m - j\mathbb{1}/q_i)}{g(-j\mathbb{1}/q_i)} \right) \prod_{j=1}^{\delta} \frac{g(c_j)g(-c_j)}{g(m + c_j)g(-m - c_j)} \ .$$

Using Hasse-Davenport, the first product can be rewritten as

$$\prod_{i=1}^{r} -\omega(p_i)^{-p_i m} g(p_i m) \ .$$

Similarly, the second product is equal to

$$\prod_{i=1}^{s} -\omega(q_i)^{q_i m} g(-q_i m) \ .$$

To rewrite the third product, we use the evaluation $g(n)g(-n) = \omega(-1)^n q$ if $n \not\equiv 0 \pmod{\mathbb{1}}$ and $g(0)g(0) = \omega(-1)^0$. We get

$$\omega(-1)^{\delta m} q^{-s(0)+s(m)} \ .$$

Finally notice that $d + \delta$ equals the degree of $\prod_{i=1}^{s}(X^{q_i} - 1)$, in particular $d + \delta = \sum_{i=1}^{s} q_i$. Hence the coefficient (without the factor $1/(1-q)$) of $\omega(t)^m$ becomes

$$(-1)^{r+s} q^{-s(0)+s(m)} \prod_{i=1}^{r} g(p_i m) \prod_{j=1}^{s} g(-q_j m) \omega \left((-1)^{\sum_i q_i} \frac{\prod_i q_i^{q_i}}{\prod_j p_j^{p_j}} \right)^m \ ,$$

as asserted.

□

As we noted in the introduction the new summation makes sense for any choice of q relatively prime with the common denominator of the α_i, β_j. We now continue to work with H_q as defined by the new summation.

Before we proceed we can now be more specific about the arithmetic nature of H_q. We need our main Theorem 1.5 for this, which we will prove in Section 5.

Theorem 4.2 *Suppose H_q is defined over $\mathbb{Q}$. Define*

$$\lambda = \min_{x\in[0,1]} \{p_1 x\} + \cdots + \{p_r x\} + \{-q_1 x\} + \cdots + \{-q_s x\} \ ,$$

where as usual $\{x\}$ denotes the fractional part of x. Then $\lambda \geq 1$ and $q^{\min(r,s)-\lambda} H_q(\boldsymbol{\alpha}, \boldsymbol{\beta}|t) \in \mathbb{Z}$.

In most cases it turns out that $\lambda - \min(r,s)$ is the exact p-adic valuation of H_q. but we have not tried to prove this.

Proof : From Theorem 1.5 it follows that the values of H_q have a denominator dividing $p^{\min(r,s)-1}$. To determine the p-adic valuation of H_q we use Theorem 2.4 and the notation therein. To be more precise, we determine the $\mathfrak{p}$-adic valuation of each coefficient $g(p_1 m)\cdots g(p_r m) g(-q_1 m)\cdots g(-q_s m)$. Because $p_1 + \cdots p_r - q_1 - \cdots - q_s = 0$ each such coefficient is in $\mathbb{Q}(\zeta_{q-1})$. According to Theorem 2.4 its $\mathfrak{p}$-adic valuation is given by

$$\sum_{i=1}^{f} \left\{ \frac{p_1 m}{q-1} \right\} + \cdots + \left\{ \frac{p_r m}{q-1} \right\} + \left\{ \frac{-q_1 m}{q-1} \right\} + \cdots + \left\{ \frac{-q_s m}{q-1} \right\} \ .$$

This is bounded below by $f\lambda$. Hence the $\mathfrak{p}$-adic valuation of H_q is bounded below by $f\lambda$, and since the values of H_q are in $\mathbb{Q}$ our theorem follows.

□

The main purpose of this paper is to show that the Katz hypergeometric function occurs naturally in the point counting numbers of the completion and normalization of the variety V_λ, which is defined by the points of $\mathbb{P}^{r+s-1}$ satisfying the equations

$$x_1 + x_2 + \cdots + x_r - y_1 - \cdots - y_s = 0, \qquad \lambda x_1^{p_1} \cdots x_r^{p_r} = y_1^{q_1} \cdots y_s^{q_s}$$

with $\lambda \in \mathbb{F}_q^\times$.

We have the following preliminary result.

Proposition 4.3 *Assume that* $\gcd(p_1, \ldots, p_r, q_1, \ldots, q_s) = 1$. *Let $V_\lambda(\mathbb{F}_q^\times)$ be the set of points on V_λ with coordinates in $\mathbb{F}_q^\times$. Then*

$$|V_\lambda(\mathbb{F}_q^\times)| = \frac{1}{q}(q-1)^{r+s-2} + \frac{1}{q(q-1)} \sum_{m=0}^{q-2} g(p_1 m)\cdots g(p_r m) g(-q_1 m)\cdots g(-q_s m) \omega(\epsilon\lambda)^m \ ,$$

where $\epsilon = (-1)^{\sum_i q_i}$.

Proof : We will use Lemma 2.7. Let us take $n+1 = r+s$ and write $\mathbf{a} = (a_1, a_2, \ldots, a_{n+1}) = (p_1, \ldots, p_r, -q_1, \ldots, -q_s)$. In particular $n = r+s-1$. We introduce the new variables $x_{r+j} = -y_j$ for $j = 1, \ldots, s$. The equations for V_λ now read

$$x_1 + x_2 + \cdots + x_{n+1} = 0, \qquad \lambda x_1^{a_1} \cdots x_{n+1}^{a_{n+1}} = \epsilon \ .$$

In vector notation the latter equation reads $\lambda \mathbf{x}^{\mathbf{a}} = \epsilon$. We dehomogenize by setting $x_{n+1} = 1$. Then

$$q|V_\lambda(\mathbb{F}_q^\times)| = \sum_{v\in\mathbb{F}_q, \mathbf{x}\in(\mathbb{F}_q^\times)^n, \lambda\mathbf{x}^{\mathbf{a}}=\epsilon} \psi_q(v(1+x_1+\cdots+x_n)) \ .$$

This is a function of $\lambda \in \mathbb{F}_q^\times$. Let $\sum_{m=0}^{q-2} c_m \omega(\lambda)^m$ be its Fourier series. The coefficient c_m is given by

$$c_m = \frac{1}{q-1} \sum_{v\in\mathbb{F}_q, \lambda, x_i\in\mathbb{F}_q^\times, \lambda\mathbf{x}^{\mathbf{a}}=\epsilon} \psi_q(v(1+x_1+\cdots+x_n))\ \omega(\lambda)^{-m} \ .$$

Substitute $\lambda = \epsilon \mathbf{x}^{-\mathbf{a}}$ to get

$$c_m = \frac{1}{q-1} \sum_{v\in\mathbb{F}_q, x_i\in\mathbb{F}_q^\times} \psi_q(v(1+x_1+\cdots+x_n))\ \omega(\epsilon\mathbf{x}^{\mathbf{a}})^m \ .$$

Then Lemma 2.7, using $\gcd(a_1,\ldots,a_n) = 1$, yields

$$c_m = (q-1)^{r+s-2}\delta(m) + \frac{1}{q-1}\prod_{i=1}^{r+s} g(a_i m)\omega(\epsilon)^m \ .$$

Putting everything together and dividing by q gives us

$$|V_\lambda(\mathbb{F}_q^\times)| = \frac{1}{q}(q-1)^{r+s-2} + \frac{1}{q(q-1)}\sum_{m=0}^{q-2} g(a_1 m)\cdots g(a_{r+s}m)\omega(\epsilon\lambda)^m \ ,$$

which yields the desired statement.

□

Notice that most terms in the summation in Proposition 4.3 and in the summation of Theorem 1.3 agree except for a few, which differ by a factor which is a power of q. The final goal of this paper is to show that this difference is caused by the difference between V_λ and its completion. In others words, the extra factors q arise from addition of the components of the completion of V_λ. In the next section we shall elaborate on this idea and prove our main Theorem 1.5.

5 Proof of the main theorem

Let notation be as in the introduction. We slightly rewrite the equations in Theorem 1.5. Introduce the variables $x_{r+j} = -y_j$ for $j = 1,\ldots,s$ and write $\mathbf{a} =$

$(a_1, a_2, \ldots, a_{r+s}) = (p_1, \ldots, p_r, -q_1, \ldots, -q_s)$. Letting $k = r + s$ our equations can be rewritten as

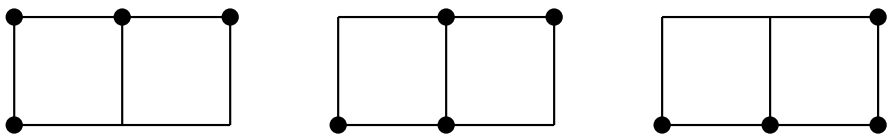
$$\sum_{i=1}^{k} x_i = 0, \qquad \lambda \prod_{i=1}^{k} x_i^{a_i} = \epsilon \ ,$$

where $\epsilon = (-1)^{\sum_i q_i}$. Consider the real $(k-1)$-dimensional cone $\mathcal{C}$ defined by $\sum_{i=1}^{k} a_i \xi_i = 0$ and $\xi_i \geq 0$ for all i. Note that it contains the vectors $\mathbf{v}_{ij}$, $i = 1, \ldots, r; j = 1, \ldots, s$ defined by $(\mathbf{v}_{ij})_i = 1/p_i, (\mathbf{v}_{ij})_{r+j} = 1/q_j$ and all other coordinates zero. The vectors $\mathbf{v}_{ij}$ span the cone $\mathcal{C}$ over $\mathbb{R}_{\geq 0}$. Let Δ^{r-1} be the standard simplex defined by $p_1\xi_1 + \cdots + p_r\xi_r = 1$ and $\xi_i \geq 0$ and Δ^{s-1} the standard simplex defined by $q_1\xi_{r+1} + \cdots + q_s\xi_{r+s} = 1,\ \xi_{r+1}, \ldots, \xi_{r+s} \geq 0$. Then we see that the points corresponding to the vectors $\mathbf{v}_{ij}$ are vertices of a polytope isomorphic to $\Delta^{r-1} \times \Delta^{s-1}$. Thus $\mathcal{C}$ is a cone over the $(k-2)$-dimensional product simplex $\Delta^{r-1} \times \Delta^{s-1}$. A *simplicial subcone* of $\mathcal{C}$ is the $\mathbb{R}_{\geq 0}$-span of any set of vectors $\mathbf{v}_{i_1 j_1}, \ldots, \mathbf{v}_{i_{k-1} j_{k-1}}$ such that $\{i_1, \ldots, i_{k-1}\} = \{1, \ldots, r\}$ and $\{j_1, \ldots, j_{k-1}\} = \{1, \ldots, s\}$. By abuse of language we will call the corresponding set of index pairs $(i_1, j_1), \ldots, (i_{k-1}, j_{k-1})$ a *simplex*. To describe a completion of V_λ we need a triangulation of $\mathcal{C}$. That is, we have to write $\mathcal{C}$ as a union of simplicial cones whose interiors do not intersect. We shall use the following result.

Proposition 5.1 *A triangulation of $\mathcal{C}$ is given by the set of cones over all simplices $(i_1, j_1), \ldots, (i_{k-1}, j_{k-1})$ with $i_1 \leq i_2 \leq \cdots \leq i_{k-1}$ and $j_1 \leq j_2 \leq \cdots \leq j_{k-1}$. We call this the standard triangulation.*

Thus simplices of the standard triangulation are in one-to-one correspondence with lattice paths that go from $(1, 1)$ to (r, s) by making steps of size 1 in north or east direction.

To illustrate this proposition we give the standard triangulation for $\Delta^2 \times \Delta^1$, which is a triangular prism. In this case $r = 3$, $s = 2$, and $k = 5$. The sequences $(i_1, j_1), \ldots, (i_4, j_4)$ are depicted as dots in the following pictures,



There are three possible pictures which corresponds to the fact that a triangulation of a triangular prism consists of three simplices.

Any subset of a simplex is called a *cell*. The cone spanned by the corresponding vectors $\mathbf{v}_{ij}$ is called a *cellular cone*. From now on we restrict to cells which are subset of the simplices belonging to the standard triangulation. They are characterized by sequences $(i_1, j_1), \ldots, (i_l, j_l)$ with $i_1 \leq \cdots \leq i_l$ and $j_1 \leq \ldots \leq j_l$, but $\{i_1, \ldots, i_l\}$ need not equal $\{1, 2, \ldots, r\}$ anymore and similarly for $\{j_1, \ldots, j_l\}$. These cells have dimension $l - 1$ and can be pictured in a similar way as above.

The set of cellular cones is a fan. Thus it gives rise to a toric variety X. The variety V_λ is naturally a hypersurface in the open dense torus of X, so we construct our completion of V_λ by taking its closure $\overline{V_\lambda}$ in X. Note that X can be naturally decomposed into a union of components indexed by the cells. This induces a decomposition of the completion $\overline{V_\lambda}$. Below we give an explicit construction of the components of $\overline{V_\lambda}$ indexed by the cells. This construction is suitable for counting points.

Choose a cell C given by $(i_1, j_1), \ldots, (i_l, j_l)$. To it we shall associate a component of the completion of V_λ. We do this by induction on l. Let S be the support of the vectors $\mathbf{v}_{i_1 j_1}, \ldots, \mathbf{v}_{i_l j_l}$. Notice that $l + 1 \leq |S| \leq \max(k, 2l)$. We shall suppose that $|S| \leq k - 2$, when $|S| > k - 2$ there are no more components. For the construction of a component we choose an index $\sigma \notin S$ and set $x_\sigma = 1$ (dehomogenization). By induction on l we construct a component, denoted by $W(C, \lambda)$, which is given by the product of $\mathbb{G}_m^{|S|-l-1}$ times the variety given by the equations

$$1 + \sum_{i \notin S \cup \sigma} x_i = 0, \quad \lambda z^{a_S} \prod_{i \notin S \cup \sigma} x_i^{a_i} = \epsilon \tag{1}$$

in the space $(\mathbb{C}^\times)^{k-|S|}$ with coordinates $z, \{x_i\}_{i \notin S \cup \sigma}$, and where $a_S = \gcd_{i \in S}(a_i)$. When $l = 0$, the case of the empty cell, we have the equations of V_λ, namely $1 + \sum_{i \neq \sigma} x_i = 0$ and $\lambda \prod_{i \neq \sigma} x_i^{a_i} = \epsilon$ in the space $(\mathbb{C}^\times)^{k-1}$.

We start the induction with $l = 1$. In that case C is given by (i, j), say, and $S = \{i, r + j\}$. Let $b = q_j/a_S, b' = p_i/a_S$, where now $a_S = \gcd(a_i, a_{r+j}) = \gcd(p_i, q_j)$. Let $c, c' \in \mathbb{Z}$ be such that $b'c - bc' = 1$ and introduce new variables z, z' by $x_i = (z')^b z^c, x_{i+j} = (z')^{b'} z^{c'}$. Substitute this in the equations for V_λ. Then the variable z' cancels out in the second equation. We set $z' = 0$ and we are left with the equations (1) in the case $C = (i, j)$.

Now let $l \geq 1$ and suppose that we have constructed the components corresponding to length l cells with the appropriate equations. Let C be such a cell and extend it with the point (i, j) to get a cell of length $l + 1$. Denote the support of C by S and the support of the extended cell by S'. There are three possibilities, $i \in S, j \notin S$ or $i \notin S, j \in S$ or $i, j \notin S$. Let us deal with the first case. Then $S' = S \cup \{r + j\}$. The quantity $|S| - l - 1$ does not change. Note that $a_{S'} = \gcd(a_S, a_{r+j}) = \gcd(a_S, q_j)$. Let $b = q_j/a_{S'}, b' = a_S/a_{S'}$. Choose integers c, c' such that $b'c - bc' = 1$ and introduce new variables z, z' by $z := (z')^b z^c, x_{r+j} = (z')^{b'} z^{c'}$. Substitute these in (1). The variable z' cancels from the second equation. We set $z' = 0$ and arrive at the equations for $W(C \cup (i, j), \lambda)$.

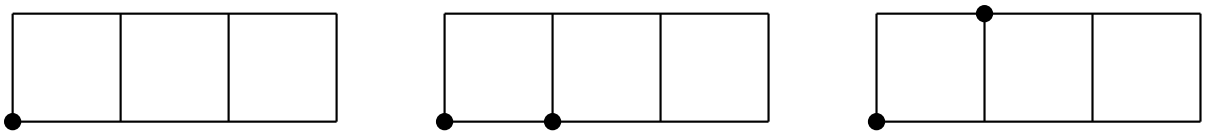
The case $i \notin S, j \in S$ is similar, so let us turn to $i, j \notin S$. Then $S' = S \cup \{i, r + j\}$ and $|S| - l - 1$ increases by 1. Let $b = q_j/a_{i,r+j}, b' = p_i/a_{i,r+j}$, where $a_{i,r+j} = \gcd(a_i, a_{r+j}) = \gcd(p_i, q_j)$. Let $c, c' \in \mathbb{Z}$ be such that $b'c - bc' = 1$ and introduce new variables z', z'' by $x_i = (z'')^b (z')^c, x_{i+j} = (z'')^{b'} (z')^{c'}$. Substitute this in (1). Then the variable z'' cancels out in the second equation. We set $z'' = 0$ and we are left with

$$1 + \sum_{i \notin S' \cup \sigma} = 0, \quad \lambda (z')^{a_{i,r+j}} z^{a_S} \prod_{i \notin S' \cup \sigma} x_i^{a_i} = \epsilon \, .$$

Notice that $a_{S'} = \gcd(a_S, a_{i,r+j})$. Let $b = a_S/a_{S'}, b' = a_{i,r+j}/a_{S'}$ and choose integers c, c' such that $b'c - bc' = 1$. Make the substitutions $z := (z')^{-b} z^{-c}, z' := (z')^{b'} z^{c'}$ and we see that z' disappears from both equations and we are left with the equations for $W(C \cup (i,j), \lambda)$. The variable z' can be chosen arbitrarily, which adds a factor $\mathbb{G}_m$. Before giving the point count on $W_{C,\lambda}$ we present an example of the evaluation of some components. Consider the case $p_1 = 6, p_2 = 3, p_3 = 2, p_4 = 1, q_1 = 8, q_2 = q_3 = 4$. We have $r = 4, s = 2$ and the equations of V_λ read

$$x_1 + x_2 + x_3 + x_4 = y_1 + y_2, \quad \lambda x_1^6 x_2^3 x_3^2 x_4 = y_1^8 y_2^4 \, .$$

We would like to construct the components $W_{C,\lambda}$ for the following cells C:



In all three examples we shall set $x_4 = 1$. First consider the cell given by $(1,1)$. We substitute $x_1 = (z')^2 z, y_1 = (z')^3 z$. This gives us, after cancellation of z' in the second equation,

$$(z')^2 z + x_2 + x_3 + 1 = (z')^3 z_2 + y_2, \quad \lambda x_2^3 x_3^3 = z^2 y_2^4 \, .$$

Now set $z' = 0$ to get the component given by

$$x_2 + x_3 + 1 = y_2, \quad \lambda x_2^3 x_3^3 = z^2 y_2^4 \, .$$

Now consider the cell given by $(1,1), (2,1)$. It means we let x_2 go to 0. Substitute $x_2 = (z')^2 z, z := (z')^3 z^2$ in the equations above,

$$(z')^2 z + x_3 + 1 = y_2, \quad \lambda x_3^3 = z y_2^4 \, .$$

Set $z' = 0$ and we get $x_3 + 1 = y_2, \lambda x_3^3 = z y_2^4$.
Now consider the cell given by $(1,1), (2,2)$. That means we will let x_2 and y_2 go to 0. In the equations for $W((1,1), \lambda)$ we make the substitution $x_2 = (z'')^4 z', y_2 = (z'')^3 z'$. We get

$$(z'')^4 z' + x_3 + 1 = (z'')^3 z', \quad \lambda x_3^3 = z^2 z' \, .$$

Set $z'' = 0$ and we get the component given by $x_3 + 1 = 0, \lambda x_3^3 = z^2 z'$. Make the substitution $z := z', z' := z(z')^{-2}$ and we get $x_3 + 1 = 0, \lambda x_3^3 = z$. The coordinate z' has cancelled from the equations, and we can choose it arbitrarily. This gives us a factor $\mathbb{G}_m$.

Proposition 5.2 *Let C be a cell as above, l its number of elements and S its support. Let $W_{C,\lambda}$ be the product of $\mathbb{G}_m^{|S|-l-1}$ and the variety defined by the equations (1). Then*

$$|W_{C,\lambda}(\mathbb{F}_q^\times)| = \frac{1}{q}(q-1)^{k-l-2} + \frac{(-1)^{|S|}}{q}(q-1)^{|S|-l-1} \sum_{m=0}^{q-2} \delta(a_S m) \left(\prod_{i=1}^{k} g(a_i m) \right) \omega(\epsilon\lambda)^m \, .$$

In addition, if we take C empty, then $|S| = l = 0$ and the point count coincides with Proposition 4.3 by setting $a_S = 0$.

Proof : We work with a non-empty cell. First we count the number N of solutions of equations (1) and then multiply the result by $(q-1)^{|S|-l-1}$. Note that

$$qN = \sum_{z,x_i \in \mathbb{F}_q^\times, v \in \mathbb{F}_q, \epsilon = \lambda z^{a_S} \prod_i x_i^{a_i}} \psi_q(v(1+\sum_i x_i)) \ ,$$

where the summation and product over i are meant to be taken over all $i \notin S$ and $i \neq \sigma$. We determine the Fourier coefficient c_m of this expression.

$$c_m = \frac{1}{q-1} \sum_{z,x_i,\lambda \in \mathbb{F}_q^\times, v \in \mathbb{F}_q, \epsilon = \lambda z^{a_S} \prod_i x_i^{a_i}} \psi_q(v(1+\sum_i x_i)) \ \omega(\lambda)^{-m} \ .$$

Substitute $\lambda = \epsilon z^{-a_S} \prod_i x_i^{-a_i}$ to get

$$c_m = \frac{1}{q-1} \sum_{z,x_i \in \mathbb{F}_q^\times, v \in \mathbb{F}_q} \psi_q(v(1+\sum_i x_i)) \ \omega(z^{a_S} \prod_i x_i^{a_i})^m \omega(\epsilon)^m \ .$$

Summation over z yields the factor $(q-1)\delta(a_S m)$. Then, using Lemma 2.7, summation over v and the x_i yields

$$c_m = \delta(a_S m) \left((q-1)^{k-|S|-1} \delta(am) + \frac{1}{q-1} g(-\sum_i a_i m) \prod_i g(a_i m) \right) \omega(\epsilon)^m \ ,$$

where $a = \gcd_{i \notin S \cup \sigma}(a_i)$. Note that any divisor of both a_S and a would also have to divide a_σ because of $\sum_{i=1}^k a_i = 0$. But since $\gcd(a_1, \ldots, a_k) = 1$ this common divisor can only be 1. Hence $\delta(a_S m)\delta(am) = \delta(m)$. Secondly, notice that if $a_S m \equiv 0 \pmod{\mathbb{1}}$ then $a_i m \equiv 0 \pmod{\mathbb{1}}$, hence $g(a_i m) = -1$, for all $i \in S$, and $-\sum_i a_i m \equiv -\sum_{i=1}^k a_i m + a_\sigma m \pmod{\mathbb{1}} \equiv a_\sigma m \pmod{\mathbb{1}}$. Hence

$$\delta(a_S m) g(-\sum_i a_i m) \prod_i g(a_i m) = (-1)^{|S|} \delta(a_S m) \prod_{i=1}^k g(a_i m) \ .$$

We obtain

$$c_m = (q-1)^{k-|S|-1} \delta(m) + \delta(a_S m) \prod_{i=1}^k g(a_i m) \ .$$

Division by q, multiplication by $(q-1)^{|S|-l-1} \omega(\lambda)^m$, and summation over m yields the desired result.

□

Denote by $\mathcal{T}_{rs}$ the set of all the cells of the standard triangulation of $\Delta^{r-1} \times \Delta^{s-1}$. Recall that any such cell C of $\mathcal{T}_{rs}$ is given by a sequence $(i_1, j_1), \ldots, (i_l, j_l)$ of distinct pairs such that $i_{m+1} - i_m, j_{m+1} - j_m \geq 0$ for $m = 1, \ldots, l-1$. The number $l = l(C)$

is called the length of the cell. As before, the support $S(C)$ is the set of indices $i_1, \ldots, i_l, r+j_1, \ldots, r+j_l$. Notice that the difference $|S(C)| - l(C) - 1$ is equal to the number of indices m such that $i_{m+1} - i_m, j_{m+1} - j_m > 0$. We note specifically that the sequence may be empty, i.e. $l = 0$, in which case we speak of the empty cell.

Proposition 5.3 *With the above notation we have*

$$\sum_C (q-1)^{|S(C)|-l(C)}(-1)^{|S(C)|} = q^{\min(r,s)} ,$$

where the summation extends over all cells C of $\mathcal{T}_{rs}$, including the empty one.

Proof : We divide the cells into two types. The set A of cells for which there exists m such that $i_m \neq j_m$ and the set B of cells for which $i_m = j_m$ for all m, including the empty one. We show that the total contribution in the sum coming from the cells in A is zero. Denote by p the minimum of all $\min(i_m, j_m)$ for which $i_m \neq j_m$. We distinguish two types in the set A.

(i) The point (p, p) is contained in the cell.

(ii) The point (p, p) is not contained in the cell.

We can map the cells from $A(i)$ one-to-one to $A(ii)$ by deletion of the point (p, p). In the process the difference $|S(C)| - l(C)$ does not change but the value of $|S(C)|$ changes by one. Hence the terms that are paired by this mapping cancel. So it remains to compute the contribution from the cells of B. They are all of the form $(i_1, i_1), \ldots, (i_l, i_l)$ and the length is at most $\min(r, s)$. For each cell in B the number $|S| = 2l$ is even and also, $|S| - l = l$. The number of cells in B of length l equals $\binom{\min(r,s)}{l}$. Hence the sum reads

$$\sum_{l=0}^{\min(r,s)} \binom{\min(r,s)}{l} (q-1)^l = (q-1+1)^{\min(r,s)} = q^{\min(r,s)} ,$$

as asserted.

□

Proposition 5.4 *Let $k = r + s$. With the above notation we have*

$$\sum_C (q-1)^{k-l(C)-1} = \sum_{m=0}^{\min(r-1,s-1)} \binom{r-1}{m}\binom{s-1}{m} q^{k-m-1} ,$$

where the summation extends over all cells C of $\mathcal{T}_{rs}$, including the empty one.

Proof : We apply induction with respect to r. Denote the sum on the left by A_{rs}. When $r = 1$ the number of cells of length l is equal to $\binom{s}{l}$. Hence

$$A_{1s} = \sum_{l=0}^{s} \binom{s}{l} (q-1)^{1+s-l-1} = q^s \ ,$$

which is the desired answer when $r = 1$. Similarly we have the formula $A_{r1} = q^r$. Now suppose that $r > 1$ and we make the induction step as follows. We claim that for all $r, s > 1$,

$$A_{rs} = qA_{r-1,s} + qA_{r,s-1} - (q^2 - q)A_{r-1,s-1} \ .$$

The summation over the cells of $\mathcal{T}_{rs}$ consists of the cells in $\mathcal{T}_{r-1,s}$, the cells $\mathcal{T}_{r,s-1}$ and the cells with $(i_l, j_l) = (r, s)$. The contribution to A_{rs} from $\mathcal{T}_{r-1,s}$ equals $(q-1)A_{r-1,s}$ because k is increased by 1. Similarly we get a contribution from $\mathcal{T}_{r,s-1}$. However, cells in $\mathcal{T}_{r-1,s-1}$ have been counted doubly. So we have to subtract the latter contribution once, which equals $(q-1)^2 A_{r-1,s-1}$. This gives the contribution

$$(q-1)A_{r-1,s} + (q-1)A_{r,s-1} - (q-1)^2 A_{r-1,s-1} \ .$$

The cells with endpoint (r, s) have not been counted yet. They arise by adding (r, s) to the cells in $\mathcal{T}_{r-1,s-1}, \mathcal{T}_{r-1,s}$ and $\mathcal{T}_{r,s-1}$. The length of the cell increases by 1 and k increases by 1, except with $\mathcal{T}_{r-1,s-1}$ in which case k increases by 2. Using inclusion-exclusion we arrive at a contribution

$$A_{r-1,s} + A_{r,s-1} - (q-1)A_{r-1,s-1} \ .$$

Our claim now follows. We see that the A_{rs} are uniquely determined by the recursion and $A_{r1} = q^r, A_{1s} = q^s$. It remains to verify that the right hand side of our proposition satisfies the same properties. We verified this for $r = 1$ and $s = 1$. Now take $r, s > 1$ and consider the right hand side

$$B_{rs} = \sum_{m \geq 0} \binom{r-1}{m} \binom{s-1}{m} q^{k-m-1} \ ,$$

where we sum over $m \geq 0$ and use the convention that $\binom{a}{m} = 0$ if $m > a$. We get that $B_{rs} - qB_{r-1,s} - qB_{r,s-1} + (q^2 - q)B_{r-1,s-1}$ equals

$$\sum_{m \geq 0} \left(\binom{r-1}{m}\binom{s-1}{m} - \binom{r-2}{m}\binom{s-1}{m} - \binom{r-1}{m}\binom{s-2}{m} + \binom{r-2}{m}\binom{s-2}{m} \right) q^{k-m-1}$$
$$- \binom{r-2}{m}\binom{s-2}{m} q^{k-m-2}.$$

The terms of the sum on the first line are easily seen to equal $\binom{r-2}{m-1}\binom{s-2}{m-1}q^{k-m-1}$. The latter is 0 if $m = 0$. So we are left with

$$\sum_{m \geq 1} \binom{r-2}{m-1}\binom{s-2}{m-1} q^{k-m-1} - \sum_{m \geq 0} \binom{r-2}{m}\binom{s-2}{m} q^{k-m-2} \ ,$$

which equals 0. Hence B_{rs} satisfies the recurrence and the equality $A_{rs} = B_{rs}$ holds for all r, s.

□

Proposition 5.5 *Let $k = r + s$. With the above notation we have*

$$\sum_{S(C)=k} (q-1)^{k-l(C)-1} = \sum_{m=0}^{\min(r-1,s-1)} \binom{r-1}{m}\binom{s-1}{m} q^m \ ,$$

where the summation extends over all cells C of $\mathcal{T}_{rs}$ with $|S(C)| = k$.

We call cells with $|S(C)| = k$ *maximal cells*. They correspond to the simplices of our triangulation. Although these maximal cells do not contribute to the pointcount on $\overline{V}_\lambda$ it is convenient to include them in summations over the cells of $\mathcal{T}_{rs}$.
Proof : The cells C with $|S(C)| = k$ are given by $(i_1, j_1), \cdots, (i_l, j_l)$ with $(i_{m+1}, j_{m+1}) - (i_m, j_m) \in \{(1,0), (0,1), (1,1)\}$ for all m and $(i_1, j_1) = (1,1)$ and $(i_l, j_l) = (r, s)$. Let D_{rs} be the corresponding sum. We claim that

$$D_{rs} = D_{r-1,s} + D_{r,s-1} + (q-1)D_{r-1,s-1}$$

for all $r, s > 1$. This is a consequence of the fact that maximal cells in $\mathcal{T}_{rs}$ arise by adding (r, s) to a maximal cell in $\mathcal{T}_{r-1,s}$, $\mathcal{T}_{r,s-1}$ or $\mathcal{T}_{r-1,s-1}$. It is clear that $D_{1s} = D_{r1} = 1$. Together with the recurrence and an argument as in the previous proposition we arrive at our assertion.

□

According to Proposition 5.2 we associate to any cell C the counting number

$$N(C) = \frac{1}{q}(q-1)^{k-l(C)-2} + \frac{(q-1)^{|S(C)|-l(C)}}{q(q-1)} \sum_{m=0}^{q-2} (-1)^{|S(C)|} \delta(a_{S(C)}m) \left(\prod_{i=1}^{k} g(a_i m) \right) \omega(\epsilon\lambda)^k \ .$$

In particular, if $|S(C)| = l(C) = 0$ (C empty), then this number coincides with $|V_\lambda(\mathbb{F}_q^\times)|$ according to Proposition 4.3. To obtain the cardinality of $\overline{V}_\lambda(\mathbb{F}_q)$, we take the sum of $N(C)$ over all cells C with $|S(C)| \leq k-2$. A straightforward check shows that our definition gives $N(C) = 0$ if $|S(C)| = k-1$. Hence we can take the sum of $N(C)$ over all cells C and subtract the contribution from the maximal cells.
We first sum over all C and do this term by term in the above expression for $N(C)$. The summation of $\frac{1}{q}(q-1)^{k-l(C)-2}$ yields, according to Proposition 5.4,

$$\frac{1}{q-1} \sum_{m=0}^{\min(r-1,s-1)} \binom{r-1}{m}\binom{s-1}{m} q^{k-m-2} \ .$$

Now choose the m-th term in the Fourier sum. Let I be the set of i such that $a_i m \equiv 0 \pmod{\mathbb{1}}$. Suppose there are r' elements of I with $a_i > 0$ and s' elements of I such that $a_i < 0$. Since $\delta(a_{S(C)}m)$ is only non-zero if the support of C is contained in I we must sum over all such C. According to Proposition 5.3 the sum of $(-1)^{S(C)}(q-1)^{|S(C)|-l(C)}$ over such cells is equal to $q^{\min(r',s')}$. The total contribution in the sum of the $N(C)$ reads

$$\frac{1}{q-1} q^{\min(r'-1,s'-1)} g(a_1 m) g(a_2 m) \cdots g(a_k m) \ .$$

Note that $q^{\min(r',s')}$ equals $s(m)$ which we defined earlier.
We now subtract the contributions coming from the maximal cells. If C is maximal, then $\prod_{i \in S(C)} \delta(a_i m) = \prod_{i=1}^{k} \delta(a_i m)$ and the latter is 0, unless $m = 0$ in which case we get 1. So, for a maximal cell C we get

$$N(C) = \frac{(q-1)^{k-l(C)-1}}{q}((q-1)^{-1}+1) = (q-1)^{k-l(C)-2} \ .$$

According to Proposition 5.5 summation over these terms gives

$$\frac{1}{q-1} \sum_{m=0}^{\min(r-1,s-1)} \binom{r-1}{m}\binom{s-1}{m} q^m \ .$$

Together these contributions yield

$$\sum_{m=0}^{\min(r-1,s-1)} \binom{r-1}{m}\binom{s-1}{m} \frac{q^{k-m-2}-q^m}{q-1} + \frac{1}{q-1} \sum_{m=0}^{q-2} q^{s(m)-1} g(a_1 m) \cdots g(a_k m) \omega(\epsilon\lambda)^m \ .$$

Combination of this result with Theorem 1.3 yields Theorem 1.5.

6 Some alternative varieties

Notice that the dimension of the variety V_λ equals $k-3$. In some cases this is higher than expected. For example, if $\boldsymbol{\alpha} = \{1/2, 1, 2\}$ and $\boldsymbol{\beta} = \{1, 1\}$ one would get $p_1 = p_2 = 2, q_1 = q_2, q_3 = q_4 = 1$. So $k = 6$ and the dimension of V_λ equals 3. However, one would expect that the Legendre family of elliptic curves is associated to this particular choice of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. One can remedy this by noticing that the set $\{2, 2, -1, -1, -1, -1\}$ can be divided into two sets $\{2, -1, -1\}$ both of whose elements sum up to 0. In such cases the dimension of V_λ can be reduced by considering an alternative variety.
Let us write $(a_1, \ldots, a_k) = (p_1, \ldots, p_r, -q_1, \ldots, -q_s)$. Suppose $\{1, \ldots, k\}$ is a union of disjoint subsets $K_1, \ldots, K_l$ such that $\sum_{r \in K_i} a_r = 0$ for $i = 1, \ldots, l$. Let $A_i = (a_j)_{j \in K_i}$

for $i = 1, \ldots, l$. Define the variety $\mathcal{V}_\lambda$ as the subvariety of $\mathbb{P}^{|K_1|-1} \times \cdots \times \mathbb{P}^{|K_l|-1}$ given by the equations

$$\sum_{i \in K_1} x_i = \cdots = \sum_{i \in K_l} x_i = 0, \qquad \lambda x_1^{a_1} \cdots x_k^{a_k} = \epsilon \ .$$

Note that $\mathcal{V}_\lambda$ has dimension $k - 2l - 1$. In our particular example we get dimension one.

Theorem 6.1 *We use the above notation. In addition we assume that the gcd of the elements $\{a_i | i \in K_j\}$ is one for all $j = 1, \ldots, l$. Then*

$$|\mathcal{V}_\lambda(\mathbb{F}_q^\times)| = \frac{1}{q-1} \prod_{j=1}^{l} Q_{|K_j|}(q) + \frac{1}{q^l(q-1)} \sum_{m=1}^{q-2} g(a_1 m) \cdots g(a_k m) \ \omega(\epsilon\lambda)^m \ ,$$

where $Q_r(x) = ((x-1)^{r-1} + (-1)^r)/x$.

Proof : For each K_i we choose $k_i \in K_i$ and set $x_{k_i} = 1$ (dehomogenization). We introduce the short-hand notation $\Sigma_j = 1 + \sum_{i \in K_j, i \neq k_j} x_i$. Then

$$q^l |\mathcal{V}_\lambda(\mathbb{F}_q^\times)| = \sum_{u_1, \ldots, u_l \in \mathbb{F}_q, \mathbf{x} \in (\mathbb{F}_q^\times)^k, \lambda \mathbf{x}^{\mathbf{a}} = 1} \psi_q(u_1\Sigma_1 + \cdots + u_l\Sigma_l) \ .$$

The latter sum is a function of $\lambda \in \mathbb{F}_q^\times$ and the m-th coefficient of its Fourier series reads

$$c_m = \frac{1}{q-1} \sum_{u_1, \ldots, u_l \in \mathbb{F}_q, \lambda, x_i \in \mathbb{F}_q^\times, \lambda \mathbf{x}^{\mathbf{a}} = 1} \psi_q(u_1\Sigma_1 + \cdots + u_l\Sigma_l) \ \omega(\lambda)^{-m} \ .$$

In the summation over $\mathbf{x}$ we let $x_{k_i} = 1$ for $i = 1, \ldots, l$. Substitution of $\lambda = \epsilon \mathbf{x}^{-\mathbf{a}}$ yields

$$c_m = \frac{1}{q-1} \sum_{u_1, \ldots, u_l \in \mathbb{F}_q, x_i \in \mathbb{F}_q^\times} \psi_q(u_1\Sigma_1 + \cdots + u_l\Sigma_l) \ \omega(\epsilon \mathbf{x}^{\mathbf{a}})^m \ .$$

We now use Lemma 2.7 to each of the sets K_j to get

$$c_0 = \frac{q^l}{q-1} (q-1)^l \prod_{j=1}^{l} Q_{|K_j|}(q)$$

and for $m \not\equiv 0 \pmod{\mathbb{1}}$,

$$c_m = (q-1)^{l-1} g(a_1 m) \cdots g(a_k m) \ .$$

Thus we get

$$|\mathcal{V}_\lambda(\mathbb{F}_q^\times)| = \frac{1}{q-1} \prod_{j=1}^{l} Q_j(q) + \frac{1}{q^l(q-1)} \sum_{m=1}^{q-2} g(a_1 m) \cdots g(a_k m) \omega(\epsilon\lambda)^m \ ,$$

as asserted.

□

References

[1] F. Beukers, G. Heckman, Monodromy for the hypergeometric function ${}_nF_{n-1}$. Invent. Math. **95** (1989), 325–354

[2] H. Cohen, *Number Theory, Volume I: Tools and Diophantine Equations*, Graduate Texts in Math. **239**, Springer Verlag, 2007.

[3] H. Cohen, *Number Theory, Volume II: Analytic and Modern Tools*, Graduate Texts in Math. **240**, Springer Verlag, 2007.

[4] F. Rodriguez-Villegas, Integral ratios of factorials and algebraic hypergeometric functions Oberwolfach reports, July 2005, math/0701362.

[5] J. Fuselier, Hypergeometric functions over $\mathbb{F}_p$ and relations to elliptic curves and modular forms, Proc. Amer. Math. Soc. **138** (2010), 109–123.

[6] J. Greene, Hypergeometric functions over finite fields, Trans. Amer. Math. Soc. **301** (1987), 77–101.

[7] N. M. Katz, *Exponential Sums and Differential Equations*, Annals of Math Studies **124**, Princeton 1990.

[8] C. Lennon, Gaussian hypergeometric evaluations of traces of Frobenius for elliptic curves, Proc. Amer. Math. Soc. **139** (2011), 1931–1938.

[9] D. McCarthy, Transformations of well-poised hypergeometric functions over finite fields, Finite Fields and Their Applications **18** (2012), 1133–1147.

[10] K. Ono, Values of Gaussian hypergeometric series, Transactions AMS **350** (1998), 1203–1223.

[11] T. Shioda, Mordell–Weil lattices and Galois representation III, Proc. Japan Acad. **65** (1989), 300–303.